
Volatility Managed Portfolios

A Critical Assessment Across Five Asset Classes

January 2000 to December 2025

Advanced Master Project

EDHEC Business School

Volatility Management Strategies

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Abstract

This report evaluates volatility-targeting strategies on five asset classes (US equities, US long-term government bonds, commodities, USD/JPY, and the Fama-French US momentum factor) over the period January 2000 to December 2025. We construct four monthly volatility forecasts (one-month and six-month realised volatilities, GJR-GARCH(1,1) with Student- t errors, and EGARCH(1,1)) and compute the target volatility with a strictly expanding window to avoid the look-ahead bias of ?.

Sharpe ratio point estimates are economically meaningful in three cases. On the S&P 500, GJR-GARCH lifts the Sharpe ratio from 0.66 to 0.79 and shortens the longest drawdown from 52 to 40 months. On the momentum factor, the Sharpe ratio rises from 0.07 to 0.36 and the worst drawdown duration falls from 205 to 73 months. On USD/JPY, the improvement is positive but modest. Statistical inference qualifies these point estimates: using the Jobson-Korkie test with the ? correction and a block-bootstrap procedure (2,000 resamples, six-month blocks), the only asset for which the Sharpe improvement is significant at the 5% level is the momentum factor (bootstrap p -values between 0.003 and 0.015). Fama-French alpha regressions corroborate this picture: alphas on the momentum factor reach 7.5–8.7% per year with t -statistics above 2.3, whereas the equity alpha loses significance under the three- and five-factor models.

For commodities and US long-term bonds, the volatility-managed Sharpe ratio is below that of Buy and Hold even at zero transaction cost. For the S&P 500 and USD/JPY, the breakeven transaction cost is approximately 60 bps, comfortably above institutional execution costs. An equal-weighted multi-asset portfolio of the five GJR-GARCH variants achieves a higher Sharpe ratio (0.67) than any single-asset variant and reduces the maximum drawdown to 19.5%. The headline policy implication is narrower than the prior literature suggests: statistically robust evidence in favour of volatility targeting exists only for the momentum factor; for equities, the case is suggestive but not conclusive on Sharpe-ratio grounds.

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1. Theory of Volatility-Targeting Strategies

1.1. Conceptual Foundation and Formal Framework

Volatility-targeting exploits a well-documented regularity: conditional return volatility is predictable at short and medium horizons (?), while expected returns are not. ? show that scaling a portfolio weight inversely with predicted variance generates portfolios with higher Sharpe ratios and substantially reduced maximum drawdowns across US equity factors. ? extend the result to a broader cross-section of asset classes. ? dissent, arguing that the published gains largely vanish once the look-ahead bias in the target volatility is removed and once Sharpe differences are tested formally; their argument motivates two design choices in this report.

At the beginning of month t , the weight on asset i is set so that the portfolio's forecasted volatility equals a target level:

$$w_{i,t} = \frac{\sigma^{\text{target}}}{\hat{\sigma}_{i,t}}, \quad (1)$$

where σ^{target} is the long-run target volatility (annualised) and $\hat{\sigma}_{i,t}$ is the conditional one-period-ahead volatility forecast for asset i , formed at the end of month $t - 1$. The realised return of the managed portfolio is

$$r_{i,t}^{vm} = \begin{cases} r_{f,t} + w_{i,t}(r_{i,t} - r_{f,t}) & \text{long-only assets,} \\ w_{i,t} r_{i,t} & \text{long-short self-financing assets,} \end{cases} \quad (2)$$

where $r_{i,t}$ is the monthly total return of asset i and $r_{f,t}$ is the monthly risk-free rate. We follow ? in setting $k = 1$ in the more general formula $w_{i,t} = (\sigma^{\text{target}}/\hat{\sigma}_{i,t})k$; any choice $k \neq 1$ that brings realised volatility exactly onto target uses information from the entire sample and is therefore not implementable in real time. A maximum leverage constraint $w_{i,t} \leq w_{\text{max}}$ prevents unbounded leverage in extremely low-volatility regimes.

1.2. Look-Ahead Bias in the Target Volatility

The full-sample standard deviation $\sigma^{\text{full}} = \text{std}(r_1, \dots, r_T)$, used as the target volatility in ? and ?, introduces a look-ahead bias because the value of the target at any date t depends on observations from $t + 1$ to T . Following ?, we eliminate this bias by an expanding-window estimator:

$$\sigma_t^{\text{target}} = \text{std}(r_1, r_2, \dots, r_{\tau_{t-1}}), \quad (3)$$

where τ_{t-1} is the last trading day of month $t - 1$. Under this definition, σ_t^{target} is observable at time t without any future information. We impose a 48-month minimum estimation window before computing portfolio weights to ensure that the initial values of σ_t^{target} and the GARCH parameter estimates are statistically reliable. Section 5 quantifies the empirical size of the bias in our sample.

1.3. Mechanisms and Conditions for Profitability

Volatility targeting can plausibly add value through five mechanisms: the variance-risk-premium channel (?), the inverse volatility-to-return relationship (?), drawdown mitigation in crisis episodes (?), the absence of any required return-timing skill, and a reduction in negative skewness through truncation of the left tail. The strategy adds value when three conditions hold jointly: volatility must be predictable (positive Mincer-Zarnowitz R^2), the conditional Sharpe ratio must be decreasing in volatility (so that risk premia do not adequately compensate for spikes in risk), and transaction costs must be small relative to the frequency and magnitude of weight changes. If volatility is essentially unpredictable, or if the conditional Sharpe ratio is positively correlated with volatility (plausible for government bonds during flight-to-quality episodes), the strategy destroys value.

2. Data

2.1. Asset Universe and Sources

The study covers five assets representing distinct risk premia and asset classes, sourced from Bloomberg and Kenneth French's data library, spanning January 2000 to December 2025 (6,542 daily observations; 264 monthly observations after the 48-month warm-up). The Fama-French factors used in Section 6.3 are daily series aggregated to monthly returns by compounding. Table 1 summarises the asset universe.

Table 1: Asset Universe: Description and Sources

Ticker	Asset	Description	Return Type
SPXT	S&P 500 TR	S&P 500 Total Return Index	Total Return
LT09TRUU	US LT Bonds TR	Bloomberg US Long Treasury TR Index	Total Return
BCOMCOT	Bloomberg Comm.	Bloomberg Commodity Total Return Index	Total Return
USDJPY	USD/JPY	USD/JPY spot exchange rate	Price Return
MOM	US Momentum Factor	Fama-French US momentum factor	Long-short
USGG3M	3M T-Bill	Annualised yield (risk-free proxy)	Yield

We use total return indices for equities, bonds, and commodities; this is essential for a fair comparison, since the volatility-targeting weight applies to the complete economic return rather than to the price change. USD/JPY is a price return only (the carry differential is not modelled), which is a conservative treatment of FX exposure. Daily simple returns are computed as $r_t = P_t/P_{t-1} - 1$ for the four index or exchange-rate series; the momentum factor is converted from percent to decimal. The risk-free rate is compounded daily as $r_{f,t} = (1 + y_t/100)^{1/252} - 1$. Monthly returns are the cumulative product of daily returns within each calendar month.

2.2. Descriptive Statistics and Correlation Structure

Table 2: Descriptive Statistics: January 2000 to December 2025

Asset	Ann. Return	Ann. Vol.	Sharpe	Skew	Exc. Kurt
S&P 500 TR	7.8%	19.4%	0.31	−0.35	10.63
US LT Bonds TR	4.2%	6.6%	0.37	−0.01	2.64
Bloomberg Commodities	6.6%	34.0%	0.14	−0.69	9.38
USD/JPY	1.3%	9.7%	−0.04	−0.13	4.27
US Momentum Factor	3.8%	16.7%	0.12	−0.71	6.54

Three features deserve attention. Commodities have the highest volatility (34%) but the lowest Sharpe ratio among directional assets. All five assets display negative skewness and positive excess kurtosis, consistent with the fat tails commonly documented on financial return distributions. The momentum factor, despite being self-financing, shows pronounced negative skewness (−0.71) and excess kurtosis (6.54), a reflection of the momentum-crash phenomenon. Cross-asset return correlations are uniformly close to zero ($|\rho| \leq 0.02$ for every pair), as shown in Figure 1. This near-orthogonality justifies treating each asset independently and implies that the multi-asset diversification analysed in Section 6.4 stacks with within-asset volatility timing rather than substituting for it.

Asset Universe: Normalised Prices & Correlation Structure

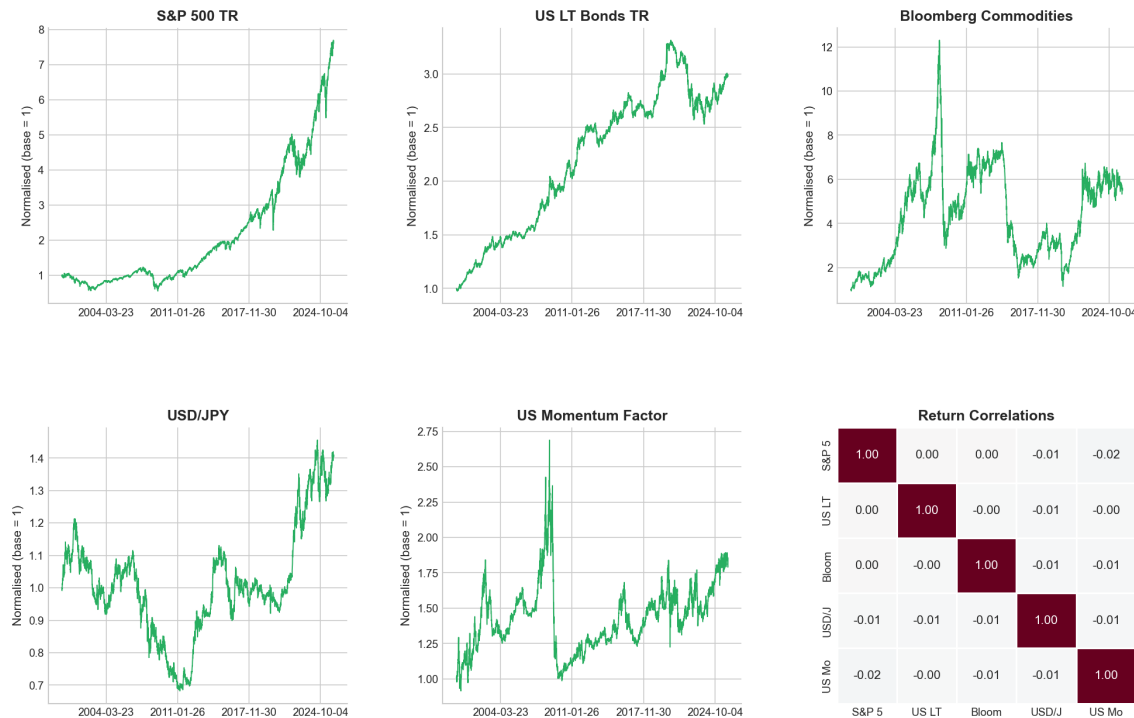


Figure 1: Normalised prices (base = 1) and pairwise return correlations across the five assets, January 2000 to December 2025.

3. Methodology

3.1. Implementation Choices

Table 3 summarises the key design choices. The 48-month warm-up ensures reliable GARCH estimation and a meaningful expanding-window target. Transaction costs are 10 bps one-way at baseline, applied as $r_t^{\text{net}} = r_t^{\text{vm}} - |\Delta w_t| \cdot c$, consistent with the institutional estimates of ?.

Table 3: Implementation Choices for the Volatility-Targeting Strategy

Parameter	Value or Rule	Rationale
Target volatility σ^{target}	Expanding window, std of daily returns up to end of month $t - 1$	No look-ahead bias (?)
Scaling constant k	1 (no ex-post rescaling)	Real-time implementable
Leverage cap w_{max}	2.0 (baseline)	Institutional constraint
Transaction cost	10 bps one-way (baseline)	?
Return frequency	Monthly (compounded from daily)	Avoids monthly-return bias
Minimum estimation window	48 months	Reliable GARCH estimation

3.2. Volatility Forecasting Models

We implement four monthly forecasts, all strictly backward-looking. The **one-month realised volatility** is the equally-weighted annualised standard deviation of daily returns in month $t - 1$: $\hat{\sigma}_{i,t}^{\text{RV1M}} = \sqrt{252/n_{t-1} \sum_{s \in t-1} r_{i,s}^2}$. The **six-month realised volatility** uses the same expression over months $t - 6$ to $t - 1$. The longer window reduces noise at the cost of slower reaction to regime shifts. The **GJR-GARCH(1,1)** of ? captures the leverage effect:

$$\sigma_{i,t}^2 = \omega + \alpha r_{i,t-1}^2 + \gamma r_{i,t-1}^2 \mathbf{1}_{[r_{i,t-1} < 0]} + \beta \sigma_{i,t-1}^2, \quad \varepsilon_{i,t} \sim t_\nu, \quad (4)$$

with $\gamma > 0$ controlling the asymmetric impact of negative shocks. The monthly forecast averages 21 daily variance forecasts. The **EGARCH(1,1)** of ? operates on log-variance, eliminating non-negativity constraints and allowing sign asymmetry independently of magnitude. Both GARCH models are re-estimated each month by quasi-maximum likelihood on an expanding daily window, with Student- t errors.

3.3. Forecast and Performance Evaluation

Forecast evaluation. We report RMSE, MAE, MAPE, the Pearson correlation between forecast and realised volatility, and the Mincer-Zarnowitz regression $\sigma_{i,t+1}^{\text{realised}} = \alpha + \beta \hat{\sigma}_{i,t} + u_{t+1}$, with the F -test of the joint null $\alpha = 0, \beta = 1$ identifying biased forecasts. Pairwise comparisons use the ? test with squared-error loss and a Newey-West variance correction.

Performance evaluation. Performance statistics are computed on monthly compounded returns and include annualised return and volatility, the Sharpe and Sortino ratios, the maximum draw-

down, the longest drawdown duration and recovery time, the 5% expected shortfall, and annualised turnover. For the S&P 500 and the momentum factor we augment the analysis with Fama-French three-factor and five-factor alpha regressions (Section 6.3).

Sharpe ratio significance. We use two complementary tests. The first is the Jobson-Korkie test with the $\hat{\rho}$ correction, valid under IID jointly normal returns:

$$z = \frac{\widehat{SR}_1 - \widehat{SR}_2}{\sqrt{\hat{\theta}/T}}, \quad \hat{\theta} = 2(1 - \hat{\rho}) + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2), \quad (5)$$

where $\widehat{SR}_j$ is the sample monthly Sharpe of strategy j and $\hat{\rho}$ the sample correlation of the two excess return series. The second is a circular block bootstrap in the spirit of $\hat{\rho}$: $B = 2,000$ resamples, six-month blocks, with the two-sided p -value computed from the bootstrap distribution centred at the observed Sharpe difference, and a 95% percentile confidence interval reported. The bootstrap is robust to autocorrelation and fat tails, both of which are present in monthly volatility-managed returns.

4. Volatility Forecast Evaluation

Table 4: Volatility Forecast Evaluation: Error Metrics, Correlation, Mincer-Zarnowitz

Asset	Model	RMSE	MAE	MAPE	Corr.	$\hat{\alpha}$	$\hat{\beta}$	R^2
S&P 500 TR	RV 1M	0.089	0.054	0.368	0.648	0.053	0.648	0.420
	RV 6M	0.103	0.061	0.441	0.479	0.060	0.559	0.229
	GJR-GARCH	0.074	0.049	0.374	0.737	0.009	0.875	0.544
	EGARCH	0.075	0.049	0.371	0.713	-0.017	1.072	0.509
US LT Bonds TR	RV 1M	0.021	0.014	0.234	0.605	0.024	0.606	0.366
	RV 6M	0.020	0.014	0.248	0.623	0.013	0.743	0.389
	GJR-GARCH	0.019	0.013	0.233	0.649	0.008	0.820	0.421
	EGARCH	0.018	0.013	0.226	0.670	0.003	0.895	0.449
Bloomberg Comm.	RV 1M	0.127	0.085	0.287	0.637	0.108	0.637	0.406
	RV 6M	0.142	0.093	0.329	0.481	0.113	0.583	0.231
	GJR-GARCH	0.123	0.088	0.326	0.620	0.043	0.779	0.385
	EGARCH	0.116	0.083	0.307	0.649	0.000	0.928	0.422
USD/JPY	RV 1M	0.037	0.027	0.312	0.514	0.042	0.514	0.264
	RV 6M	0.035	0.027	0.330	0.481	0.029	0.628	0.232
	GJR-GARCH	0.033	0.026	0.337	0.546	0.012	0.787	0.298
	EGARCH	0.033	0.025	0.325	0.532	0.005	0.879	0.283
US Momentum	RV 1M	0.054	0.038	0.303	0.810	0.025	0.810	0.656
	RV 6M	0.066	0.043	0.384	0.707	0.025	0.767	0.500
	GJR-GARCH	0.052	0.036	0.306	0.821	0.011	0.903	0.675
	EGARCH	0.052	0.036	0.307	0.817	0.012	0.890	0.668

Notes: Best performer per asset highlighted in bold. The Mincer-Zarnowitz F -test rejects the joint null $\hat{\alpha} = 0, \hat{\beta} = 1$ at $p < 0.05$ for RV 1M and RV 6M on every asset, and for GJR-GARCH on every asset except the S&P 500, where EGARCH is the only model that fails to reject ($p = 0.241$).

GJR-GARCH dominates on equities and momentum. For the S&P 500, GJR-GARCH reduces RMSE by 17.5% relative to RV 1M and raises R^2 from 0.42 to 0.54. The Mincer-Zarnowitz slope ($\hat{\beta} = 0.875$) is closer to the ideal value of one than that of RV 1M ($\hat{\beta} = 0.648$). The leverage-effect parameter $\gamma > 0$ is the likely source, as equities exhibit strong asymmetric volatility responses to negative returns.

EGARCH is the best model for bonds, commodities, and FX. EGARCH achieves the lowest RMSE for US long-term bonds (0.018), Bloomberg Commodities (0.116), and USD/JPY (0.033). Its Mincer-Zarnowitz slope is the closest to one for these three assets.

RV 6M is systematically biased. Across every asset, RV 6M displays the highest RMSE and the lowest R^2 , with Mincer-Zarnowitz slopes between 0.56 and 0.74, indicating severe underestimation during volatility spikes. Despite these deficiencies, RV 6M achieves competitive portfolio performance through its low turnover (Section 7).

Momentum is the most forecastable asset (GJR-GARCH $R^2 = 0.675$), reflecting the prolonged variance bursts that accompany momentum crashes. **USD/JPY shows the weakest forecastability** ($R^2 < 0.30$ for every model), consistent with the near-random-walk behaviour of spot exchange rates (?). Diebold-Mariano pairwise tests find no statistically significant accuracy difference between GJR-GARCH and EGARCH at the 5% level on any asset; both significantly outperform RV 1M for equities.

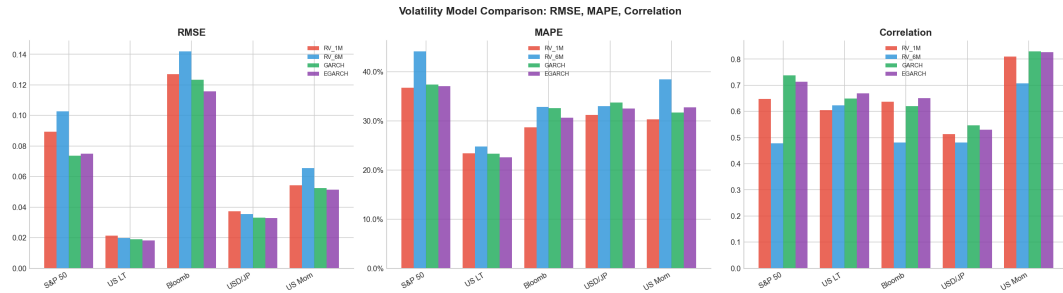


Figure 2: Volatility Model Comparison: RMSE, MAPE, and Correlation. GJR-GARCH and EGARCH dominate the realised-volatility benchmarks for equities and momentum; improvements for FX and commodities are smaller.

5. Implementation Diagnostics

5.1. Realised versus Target Volatility, and the Constant k

A volatility-targeting strategy with $k = 1$ does not, in general, deliver a realised volatility equal to the target; the ratio is informative about the unbiasedness of the volatility forecast. In our sample, the realised-to-target ratio lies between 0.94 and 1.05 for fifteen of the twenty volatility-managed entries, indicating that $k = 1$ is operationally reasonable. The largest deviations are a ratio of

0.82 for the two GARCH variants on the S&P 500 (slightly too defensive) and a ratio of 1.19 for VM RV 1M on USD/JPY (the strategy runs hot because RV 1M underestimates FX volatility). The ex-post implied scaling constant $k^* = \sigma^{\text{target}} / \text{Vol}(r/\hat{\sigma})$ is between 1.12 and 1.22 for the S&P 500 and close to one for the other assets, but it uses information from the entire sample and is therefore inadmissible for real-time deployment; we retain $k = 1$ throughout and report k^* only as a diagnostic.

5.2. Empirical Cost of the Look-Ahead Bias

To quantify the magnitude of the bias, we recompute every volatility-managed strategy under both the expanding-window estimator of Equation (3) and the look-ahead-biased full-sample estimator used by ? and ?. Table 5 reports the impact on annualised Sharpe ratios.

Table 5: Impact of the Look-Ahead Bias on Annualised Sharpe Ratios

Asset	Strategy	SR biased	SR unbiased	Δ Sharpe	Inflation
S&P 500 TR	VM RV 1M	0.781	0.809	−0.028	−3.4%
	VM GJR-GARCH	0.757	0.785	−0.027	−3.5%
US LT Bonds	VM GJR-GARCH	0.255	0.256	−0.001	−0.4%
Bloomberg Comm.	VM GJR-GARCH	0.147	0.162	−0.015	−9.2%
USD/JPY	VM GJR-GARCH	0.109	0.116	−0.007	−6.1%
US Momentum	VM GJR-GARCH	0.385	0.363	+0.022	+6.1%

The average bias-induced inflation across all twenty volatility-managed cells in our sample is −2.4%. For the S&P 500, the bias actually reduces (rather than inflates) the Sharpe ratio, the opposite of the direction emphasised by ?. The intuition is that the expanding-window target starts at its value at the end of the 48-month warm-up window, which on the S&P 500 over 2000–2003 was a high-volatility regime (the dot-com crash); the initial values of σ_t^{target} are therefore higher than the eventual full-sample standard deviation, the strategy levers up more aggressively in the early years, and the resulting Sharpe ratio is slightly higher. The largest absolute bias is on commodities and USD/JPY, the two assets for which forecast quality is the weakest. In aggregate the bias is smaller in our sample than ? document for the US equity factor universe, but only the expanding-window specification is admissible for real-time deployment.

6. Performance Evaluation

6.1. Full-Sample Performance

Table 6 reports the full-sample performance statistics for the five assets at the baseline of 10 bps transaction cost and a 200% leverage cap. Figure 3 plots the cumulative wealth paths.

Table 6: Full-Sample Performance by Asset and Strategy (TC = 10 bps, Leverage cap = 200%)

Asset	Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	DDdur (m)	Rec.(m)	ES(5%)	TO
S&P 500	Buy and Hold	10.7%	14.6%	0.658	0.871	-50.9%	52	37	-9.4%	0.00
	VM RV 1M	15.6%	17.9%	0.809	1.231	-39.3%	42	22	-10.4%	3.73
	VM RV 6M	13.4%	17.5%	0.713	1.028	-37.4%	44	24	-10.6%	1.13
	VM GJR-GARCH	14.1%	16.5%	0.785	1.193	-36.8%	40	20	-9.6%	3.88
	VM EGARCH	13.7%	16.5%	0.762	1.147	-40.6%	42	22	-9.6%	3.68
US LT Bonds	Buy and Hold	3.4%	6.4%	0.294	0.426	-22.2%	66	n/r	-3.7%	0.00
	VM RV 1M	2.9%	7.0%	0.193	0.277	-23.0%	66	n/r	-4.3%	3.06
	VM RV 6M	3.4%	6.6%	0.278	0.417	-20.9%	66	n/r	-3.9%	0.68
	VM GJR-GARCH	3.2%	6.4%	0.256	0.378	-21.1%	66	n/r	-3.8%	1.19
	VM EGARCH	3.1%	6.4%	0.236	0.346	-21.6%	66	n/r	-3.9%	1.19
Commodities	Buy and Hold	3.7%	32.9%	0.235	0.315	-88.5%	211	n/r	-21.5%	0.00
	VM RV 1M	0.8%	38.0%	0.175	0.244	-91.1%	165	n/r	-25.7%	3.50
	VM RV 6M	2.0%	34.7%	0.193	0.256	-90.2%	211	n/r	-23.6%	0.78
	VM GJR-GARCH	1.3%	32.9%	0.162	0.225	-88.3%	211	n/r	-22.1%	1.78
	VM EGARCH	1.3%	33.2%	0.162	0.227	-88.9%	211	n/r	-22.4%	1.78
USD/JPY	Buy and Hold	1.6%	9.8%	0.040	0.060	-37.1%	97	41	-6.0%	0.00
	VM RV 1M	2.0%	11.7%	0.085	0.137	-39.8%	96	35	-6.8%	3.83
	VM RV 6M	2.7%	10.9%	0.142	0.238	-36.4%	94	33	-5.9%	0.92
	VM GJR-GARCH	2.4%	10.2%	0.116	0.194	-35.4%	94	33	-5.5%	1.66
	VM EGARCH	2.2%	10.3%	0.099	0.164	-36.9%	96	35	-5.6%	1.68
Momentum	Buy and Hold	1.5%	15.0%	0.066	0.076	-61.0%	205	n/r	-11.9%	0.00
	VM RV 1M	6.6%	17.5%	0.356	0.524	-43.1%	79	70	-10.7%	2.80
	VM RV 6M	7.2%	16.5%	0.400	0.624	-35.5%	60	47	-9.6%	0.80
	VM GJR-GARCH	6.5%	16.2%	0.363	0.561	-38.4%	73	64	-9.5%	2.42
	VM EGARCH	6.3%	15.9%	0.357	0.558	-38.9%	73	60	-9.3%	2.35

Notes: n/r = not recovered at sample end (strategy still below prior peak as of December 2025).

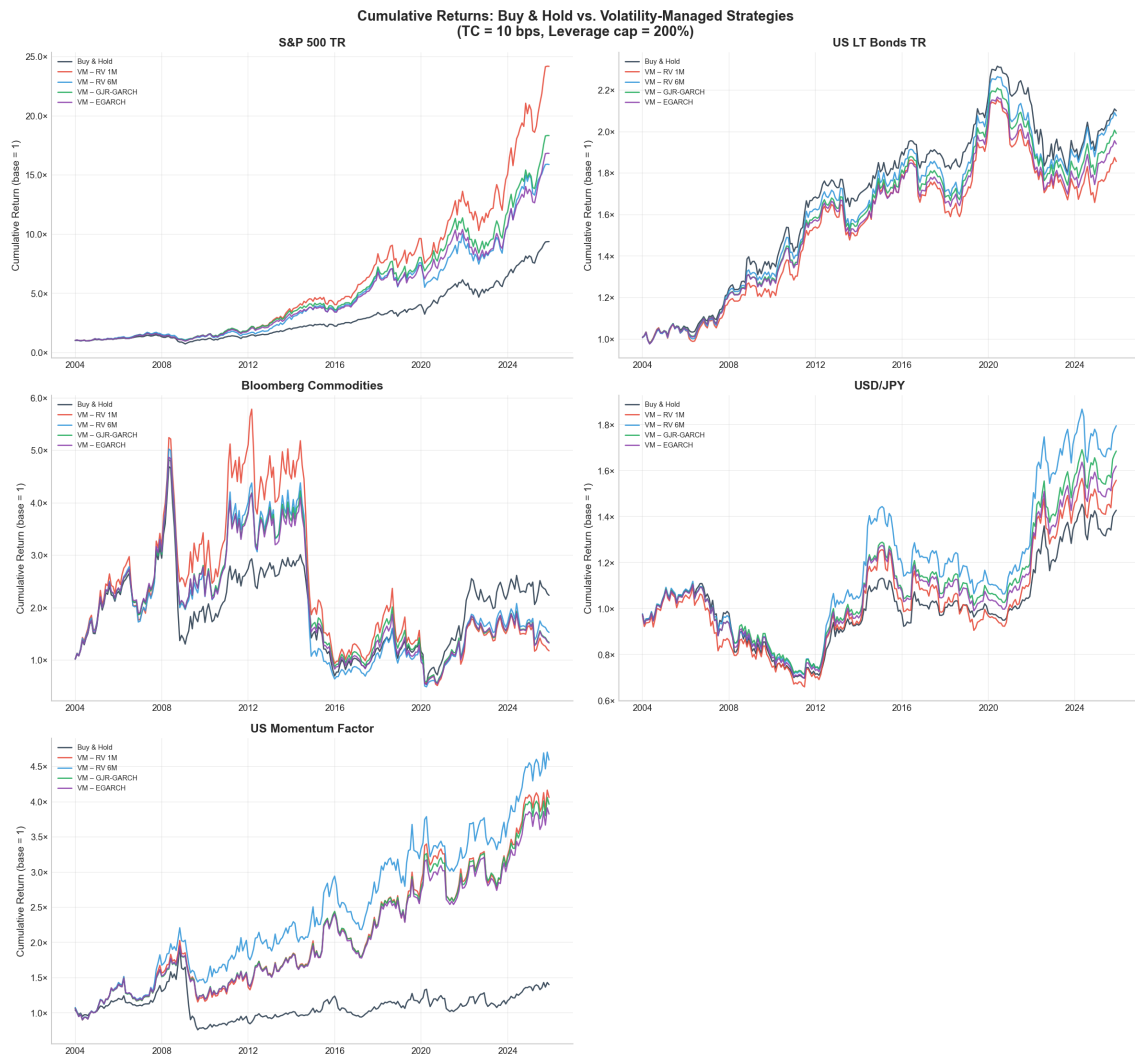


Figure 3: Cumulative Returns: Buy and Hold versus Volatility-Managed Strategies (TC = 10 bps, leverage cap = 200%).

6.2. Statistical Significance of Sharpe Ratio Differences

Table 7 reports the Jobson-Korkie-Memmel and block-bootstrap tests of every (volatility-managed strategy minus Buy and Hold) pair, and the GJR-GARCH versus EGARCH model comparison.

Table 7: Sharpe Ratio Significance Tests (VM strategies vs. Buy and Hold; selected pairs)

Asset	Pair	Δ SR	JKM z	JKM p	Boot p	Boot 95% CI
S&P 500	VM RV 1M vs. B&H	+0.151	+1.53	0.125	0.208	[−0.07, +0.38]
	VM RV 6M vs. B&H	+0.055	+0.63	0.529	0.660	[−0.18, +0.28]
	VM GJR-GARCH vs. B&H	+0.126	+1.31	0.191	0.301	[−0.10, +0.35]
	VM EGARCH vs. B&H	+0.104	+1.17	0.240	0.332	[−0.10, +0.31]
US LT Bonds	VM GJR-GARCH vs. B&H	−0.037	−0.62	0.533	0.517	[−0.15, +0.07]
Commodities	VM GJR-GARCH vs. B&H	−0.073	−1.05	0.292	0.306	[−0.21, +0.06]
USD/JPY	VM GJR-GARCH vs. B&H	+0.076	+1.42	0.155	0.192	[−0.04, +0.19]
Momentum	VM RV 1M vs. B&H	+0.290**	+2.68	0.007	0.015	[+0.06, +0.52]
	VM RV 6M vs. B&H	+0.334***	+3.15	0.002	0.003	[+0.10, +0.55]
	VM GJR-GARCH vs. B&H	+0.297**	+2.70	0.007	0.013	[+0.05, +0.53]
	VM EGARCH vs. B&H	+0.291**	+2.75	0.006	0.012	[+0.06, +0.52]

Notes: Significance $*p < 0.10$, $**p < 0.05$, $***p < 0.01$, using the more conservative of the two p -values. Bootstrap: $B = 2,000$ resamples, six-month blocks.

The headline finding is that, on conventional statistical criteria, the Sharpe ratio improvement is significant only for the momentum factor. All four momentum pairs reach $p < 0.05$ on both tests, with bootstrap confidence intervals comfortably above zero. For the other four assets, the bootstrap confidence intervals all contain zero. The S&P 500 GJR-GARCH versus Buy and Hold point estimate is +0.126 with a 95% interval of [−0.10, +0.35]: economically meaningful but not statistically distinguishable from zero at conventional levels. This pattern is exactly the one anticipated by ?. The GJR-GARCH versus EGARCH model comparison shows no statistically significant difference on any asset except US LT bonds, where GJR-GARCH outperforms by 0.020 in Sharpe; too small for practical relevance.

6.3. Fama-French Alpha Regressions

For the S&P 500 and the momentum factor, we run alpha regressions on the Fama-French three- and five-factor models:

$$r_{i,t}^{exc} = \alpha_i + \beta_i^{\text{MKT}} \text{MKT}_t + \beta_i^{\text{SMB}} \text{SMB}_t + \beta_i^{\text{HML}} \text{HML}_t \left[+ \beta_i^{\text{RMW}} \text{RMW}_t + \beta_i^{\text{CMA}} \text{CMA}_t \right] + u_{i,t}, \quad (6)$$

with Newey-West HAC standard errors (six lags). The bracketed terms are present in the five-factor specification only. For the momentum factor, we exclude the momentum factor from the right-hand side; the resulting alpha measures the value added by volatility-managing momentum on top of static equity-style exposures. We do not run Fama-French regressions for bonds, commodities, or FX because the Fama-French factors are designed to span the equity-return generating process; a proper multi-factor risk model for these assets would involve term and default factors, commodity carry and momentum factors, and FX carry and value factors. We acknowledge this as a limitation.

Table 8: Fama-French Alpha Regressions (Newey-West HAC, 6 lags)

Asset	Strategy	Model	α (ann.)	$t(\alpha)$	$p(\alpha)$	Adj. R^2
S&P 500	Buy and Hold	FF3	−0.22%	−0.98	0.328	0.997
	Buy and Hold	FF5	−0.41%**	−2.15	0.032	0.997
	VM RV 1M	FF3	+3.54%*	+1.74	0.081	0.803
	VM GJR-GARCH	FF3	+2.77%	+1.51	0.132	0.813
	VM GJR-GARCH	FF5	+2.81%	+1.58	0.115	0.815
	VM EGARCH	FF3	+2.27%	+1.39	0.163	0.838
Momentum	Buy and Hold	FF3	+3.57%	+1.22	0.224	0.179
	Buy and Hold	FF5	+2.99%	+0.94	0.348	0.211
	VM RV 1M	FF3	+8.33%**	+2.53	0.011	0.124
	VM RV 6M	FF3	+8.73%***	+2.81	0.005	0.104
	VM GJR-GARCH	FF3	+7.76%**	+2.52	0.012	0.122
	VM EGARCH	FF3	+7.49%**	+2.47	0.014	0.120

For the S&P 500, alpha point estimates are between +1.5% and +3.5% per year, but only the RV 1M alpha reaches marginal significance ($p = 0.08$). For the momentum factor, every volatility-managed alpha is between +7.5% and +8.7% with $t > 2.3$, all significant at the 5% level. The market beta on the momentum variants is consistently negative ($\hat{\beta}^{\text{MKT}} \approx -0.20$), reflecting the long-short construction, and the adjusted R^2 is low (≈ 0.12), confirming that the volatility management of momentum is a genuine timing skill rather than a tilt towards a static factor exposure.

6.4. Multi-Asset Equal-Weight Portfolio

The near-zero cross-asset correlations documented in Section 2 suggest that combining the volatility-managed strategies should deliver diversification benefits on top of within-asset timing. Table 9 reports the performance of an equal-weight portfolio of the five GJR-GARCH or EGARCH variants, rebalanced monthly, against the equal-weight Buy and Hold benchmark on the same five assets.

Table 9: Multi-Asset Equal-Weight Portfolios (TC = 10 bps)

Strategy	Ann. Ret	Ann. Vol	Sharpe	Sortino	MaxDD	ES(5%)
EW Buy and Hold	5.68%	7.52%	0.548	0.762	−25.9%	−4.56%
EW VM GJR-GARCH	7.00%	8.03%	0.673	1.026	−19.5%	−4.68%
EW VM EGARCH	6.81%	8.09%	0.646	0.984	−19.8%	−4.78%

The equal-weight GJR-GARCH portfolio achieves a Sharpe ratio of 0.673, higher than every single-asset volatility-managed Sharpe in Table 6 except for momentum (where the long-short construction is not directly comparable). The maximum drawdown is reduced by 6.4 percentage points (from 25.9% to 19.5%), while expected shortfall is essentially unchanged. Volatility timing and cross-asset diversification stack rather than substitute.

6.5. Discussion by Asset

S&P 500. GJR-GARCH raises the Sharpe ratio from 0.658 to 0.785 (a 19% increase), reduces the maximum drawdown from 50.9% to 36.8%, and cuts the recovery time from 37 to 20 months. The Sortino ratio improves from 0.871 to 1.193, indicating that the improvement comes from a reduction in downside risk. However, neither the Sharpe difference ($p = 0.30$ bootstrap, 95% CI $[-0.10, +0.35]$) nor the FF3/FF5 alpha ($p = 0.13$) is statistically significant: the economic case is suggestive, the statistical case is weaker.

US LT Bonds. Volatility management modestly destroys value (Sharpe 0.256 vs. 0.294 under GJR-GARCH). The mechanism is the flight-to-quality dynamic: bond volatility rises in equity-market crises, when bonds typically generate positive returns, so the strategy de-risks at the moments when bonds are most valuable.

Bloomberg Commodities. Volatility management is counterproductive ($\Delta SR = -0.073$). Two structural factors drive this result: the high base volatility ($\approx 33\%$) leaves little room to lever within the cap, and the catastrophic 88% drawdown from the 2008 super-cycle peak reflects fundamental price dynamics rather than the volatility-clustering shocks that GARCH-type models anticipate.

USD/JPY. A modest improvement is observed ($\Delta SR = +0.076$ for GJR-GARCH). RV 6M produces the highest Sharpe (0.142) despite GJR-GARCH dominating on forecast accuracy, because the weak FX forecastability ($R^2 < 0.30$) makes the smoother, lower-turnover RV 6M signal more economically valuable.

US Momentum Factor. The only asset on which the Sharpe improvement is both economically and statistically large. The Sharpe ratio rises from 0.066 to 0.363, the maximum drawdown shrinks from 61% to 38%, the longest drawdown shortens from 205 to 73 months, and the bootstrap p -value is 0.013. The intuition, established in ?, is that momentum crashes occur during volatility spikes (the March 2009 reversal and the March 2020 COVID shock), so the strategy avoids them almost mechanically.

7. NBER Business Cycle Analysis

A central claim of volatility-targeting proponents is that the strategy delivers automatic recession protection. Three NBER recessions overlap with our sample: the dot-com recession (March–November 2001, 9 months, which falls within the 48-month warm-up window), the Great Financial Crisis (December 2007–June 2009, 19 months), and the COVID-19 recession (February–April 2020, 3 months). After the warm-up, the usable sample contains 22 recession months out of 264, dominated by the Great Financial Crisis.

Table 10: Performance by NBER Business Cycle Phase (Annualised)

Asset	Strategy	Expansion (242 months)			Recession (22 months)		
		Ret	Vol	SR	Ret	Vol	SR
S&P 500	Buy and Hold	14.6%	12.6%	1.16	−25.4%	26.5%	−0.96
	VM GJR-GARCH	17.7%	16.3%	1.08	−18.6%	16.0%	−1.16
US LT Bonds	Buy and Hold	3.1%	6.2%	0.50	+9.0%	8.7%	+1.03
	VM GJR-GARCH	3.0%	6.4%	0.47	+6.7%	5.9%	+1.13
Commodities	Buy and Hold	15.0%	29.0%	0.52	−52.4%	59.3%	−0.88
	VM GJR-GARCH	10.9%	31.0%	0.35	−35.6%	48.6%	−0.73
USD/JPY	Buy and Hold	3.1%	9.4%	0.33	−8.7%	13.3%	−0.65
	VM GJR-GARCH	3.5%	10.2%	0.35	−4.3%	10.4%	−0.41
Momentum	Buy and Hold	3.5%	12.0%	0.29	−5.7%	34.4%	−0.17
	VM GJR-GARCH	7.5%	16.0%	0.47	+8.7%	19.5%	+0.45

Small-sample caveat. With $T = 22$ recession months, the 95% confidence interval around any single recession Sharpe is roughly ± 0.43 ; differences smaller than half a Sharpe point are not statistically distinguishable from sampling noise. With this caveat, the recession-loss reduction is meaningful for the S&P 500 (6.8 percentage points), commodities (16.8 percentage points), USD/JPY (a halving of the loss), and momentum (a sign flip from -5.7% to $+8.7\%$). For bonds, the flight-to-quality premium is partially sacrificed by the volatility-targeting adjustment, but the proportional volatility compression preserves the recession Sharpe.



Figure 4: Annualised return by NBER business cycle phase. The volatility-managed strategies provide meaningful loss reduction during recessions for the S&P 500, commodities, and momentum.

8. Transaction Costs, Turnover, and Leverage

8.1. Turnover and Transaction Cost Sensitivity

Table 11: Annualised Turnover by Asset and Model

Asset	VM RV 1M	VM RV 6M	VM GJR-GARCH	VM EGARCH
S&P 500 TR	3.73	1.13	3.88	3.68
US LT Bonds TR	3.06	0.68	1.19	1.19
Bloomberg Commodities	3.50	0.78	1.78	1.77
USD/JPY	3.83	0.92	1.66	1.68
US Momentum Factor	2.80	0.80	2.42	2.35

RV 1M generates the highest turnover (2.80–3.83), the GARCH models lie in an intermediate range (1.19–3.88), and RV 6M is consistently the lowest (0.68–1.13). At 10 bps, the cost penalty for the highest-turnover model is approximately 38 bps per year, non-trivial for a low-return asset such as USD/JPY.

Table 12 reports the breakeven transaction cost (the level at which the volatility-managed Sharpe equals Buy and Hold) under GJR-GARCH. For the S&P 500 and USD/JPY, the breakeven is comfortably above typical institutional execution costs (≈ 5 –20 bps; see ?), so the strategy retains a margin of safety. For US long-term bonds and commodities, the volatility-managed Sharpe is below Buy and Hold even at zero cost: the failure is structural, not cost-driven. On the momentum factor, the breakeven exceeds 100 bps and the volatility-managed variant is preferred at any plausible cost.

Table 12: Breakeven Transaction Cost (VM GJR-GARCH vs. Buy and Hold)

Asset	B&H Sharpe	VM Sharpe @ 0 bps	Breakeven TC
S&P 500 TR	0.658	0.808	63.6 bps
US LT Bonds TR	0.294	0.275	< 0 bps (VM never wins)
Bloomberg Commodities	0.235	0.168	< 0 bps (VM never wins)
USD/JPY	0.040	0.132	56.6 bps
US Momentum Factor	0.066	0.378	> 100 bps (VM always wins)

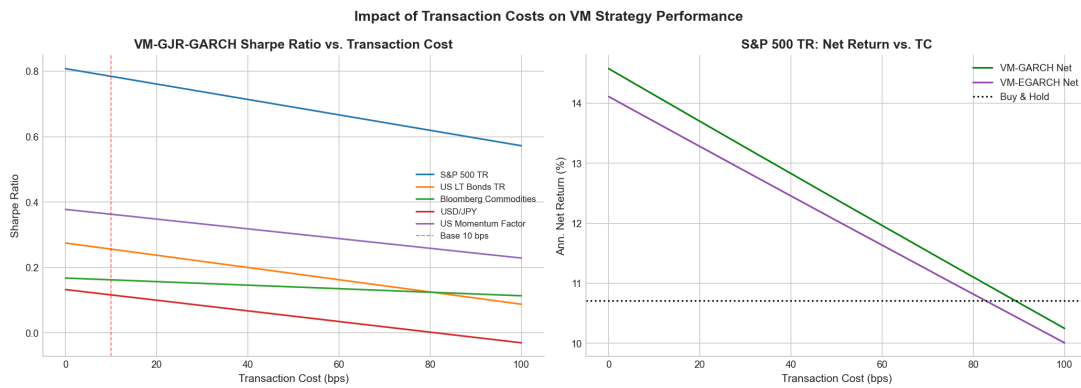


Figure 5: Impact of transaction costs on VM-GJR-GARCH performance. The breakeven cost for equities is approximately 64 bps, well above typical institutional execution costs.

8.2. Turnover Mitigation

We tested two families of turnover-reduction rules applied to VM GJR-GARCH: threshold bands (weight updated only when the raw weight differs by more than $b \in \{5\%, 10\%, 15\%, 20\%\}$ from the previous implemented weight) and exponential moving averages of the raw weights (spans $s \in \{2, 3, 6, 12\}$ months). The full grid is reported in the project notebook; Table 13 summarises the breakeven transaction cost under three rules: the baseline, the 10% band, and the EMA span = 3.

Table 13: Breakeven Transaction Cost under Mitigation Rules (basis points)

Asset	Baseline	Band 10%	EMA span 3
S&P 500 TR	63.6	65.2	> 100
US LT Bonds TR	< 0	< 0	6.3
Bloomberg Commodities	< 0	< 0	< 0
USD/JPY	56.6	66.1	> 100
US Momentum Factor	> 100	> 100	> 100

The EMA span 3 rule pushes the breakeven cost above 100 bps for the S&P 500 and USD/JPY, an improvement of approximately 60% over the baseline. For US long-term bonds, the rule moves the strategy from "never wins" to a positive (if small) breakeven, although the absolute Sharpe gap is too modest for the result to be practically relevant. Commodities remain non-viable under every mitigation rule we tested.

8.3. Leverage Sensitivity

Table 14: Leverage Constraint Sensitivity: VM GJR-GARCH Sharpe Ratios

Asset	1.0	1.25	1.5	2.0	3.0	No Cap
S&P 500 TR	0.769	0.785	0.803	0.785	0.782	0.782
US LT Bonds TR	0.289	0.281	0.259	0.256	0.256	0.256
Bloomberg Commodities	0.226	0.206	0.174	0.162	0.160	0.160
USD/JPY	0.071	0.091	0.105	0.116	0.116	0.116

For equities, the Sharpe ratio is maximised at a cap of 1.5 (0.803), marginally above the 2.0 baseline. For bonds and commodities, tighter constraints improve risk-adjusted performance monotonically: the additional leverage employed by GJR-GARCH on these assets does not generate sufficient return to compensate for its costs. For USD/JPY, the relationship is inverse and the Sharpe rises monotonically with the cap up to 2.0. In practical terms, a 1.5 cap is optimal for equities, 1.0 for bonds and commodities, and 2.0 for USD/JPY.

9. Summary and Recommendations

9.1. Synthesis

The four conditions identified in Section 1 organise the results clearly: the S&P 500, the momentum factor and US LT bonds satisfy the forecastability condition (Mincer-Zarnowitz $R^2 > 0.40$); USD/JPY does not. Forecast quality alone does not guarantee economic value, as commodities have $R^2 > 0.40$ for GJR-GARCH yet the strategy destroys value. The strategy benefits assets where the conditional Sharpe ratio is decreasing in volatility, which holds for the momentum factor in our sample but fails for US long-term bonds (flight-to-quality dynamic). The breakeven transaction cost is comfortable for equities and FX, but below zero for bonds and commodities. Critically, only the momentum factor passes the Jobson-Korkie and bootstrap Sharpe-significance tests at the 5% level; the equity improvement is suggestive but not statistically distinguishable from zero. The look-ahead bias, while smaller in our sample than ? document for US equity factors (average inflation of -2.4%), is conceptually fatal because the look-ahead-biased strategy is unimplementable. The statistical-significance issue is the more durable of the two critiques.

9.2. Recommendations to the Volatility Management Board

1. Adopt for the momentum factor. The only statistically significant improvement in the study (bootstrap $p \in [0.003, 0.015]$ for all four models, Fama-French alphas of 7.5–8.7% per year with $t > 2.3$). The point estimate is also the largest: the worst drawdown is reduced from 61% to 38% and the recovery time from a never-recovered position to 64 months. *Specification:* VM GJR-GARCH or VM RV 6M, EMA smoothing of span 3, leverage cap of 2.0.

2. Adopt for the S&P 500 with caveats. The point estimate is favourable ($\Delta SR = +0.126$, drawdown reduced from 51% to 37%, recovery time from 37 to 20 months), the breakeven cost

is comfortable (≈ 64 bps), and the recession-loss reduction is 6.8 percentage points. However, the Sharpe difference is not statistically significant ($p = 0.30$ bootstrap, 95% CI $[-0.10, +0.35]$) and the Fama-French alpha is not significant either ($p = 0.13$). The Board should adopt this as a probable but not certain Sharpe improvement. *Specification:* VM GJR-GARCH, EMA span 3 (pushes breakeven > 100 bps), leverage cap of 1.5.

3. Do not adopt for commodities. VM Sharpe below Buy and Hold at every positive cost level (breakeven < 0 bps). The structural combination of high base volatility, weak forecast quality, and the secular post-2008 commodity decline leaves the strategy without an edge. We recommend managing commodity risk through position sizing and cross-asset diversification rather than through volatility targeting.

4. Do not adopt for US long-term bonds. The flight-to-quality dynamic causes bond volatility spikes to coincide with positive return episodes; the strategy de-risks at the worst moments. The Sharpe difference is modestly negative ($\Delta SR = -0.038$) but consistent across all four models.

5. Marginal adoption for USD/JPY. The improvement is positive but modest ($\Delta SR = +0.076$ for GJR-GARCH) and not statistically significant. Given the weak FX forecastability, the strategy primarily adds value through drawdown reduction. *Specification:* VM RV 6M (lowest turnover, highest Sharpe at 0.142) or GJR-GARCH with Band 15% mitigation.

6. Model selection. Choose GJR-GARCH as the primary model on three grounds: superior RMSE and R^2 for equities and momentum, an economically interpretable leverage-effect parameter $\gamma > 0$, and lower turnover than RV 1M. EGARCH serves as a robustness check; the net Sharpe difference between the two GARCH variants is statistically insignificant on every asset except US LT bonds, where the difference is too small to be practically relevant.

7. Implementation overlays. An EMA smoothing of span 3 reduces turnover by 30–50% with minimal Sharpe sacrifice and, on equities and FX, pushes the breakeven transaction cost above 100 basis points. We recommend standardising mitigation rules across adopted strategies to simplify implementation and monitoring.

8. Multi-asset implementation. The equal-weight portfolio of the five GJR-GARCH variants achieves a Sharpe ratio (0.673) higher than any single-asset variant in our sample, with a maximum drawdown of 19.5%. Even if commodities and bonds are not adopted as standalone overlays, retaining them in a diversified multi-asset portfolio is beneficial because their volatility-timed returns are nearly orthogonal to the equity and momentum sleeves.

9.3. Limitations

The 2000–2025 sample is dominated by two major crises (2008 and 2020); the strategy's value-add may differ in a different sample window. Daily rebalancing within each month is assumed, which may not match real-world operational constraints. The transaction-cost model is simple (linear, one-way, no market-impact non-linearities). The multi-asset analysis uses equal weights only; a formal time-varying covariance optimisation is left for future work. Finally, the Fama-French risk model applies only to the S&P 500 and the momentum factor; bond, commodity, and

FX assets would require their own multi-factor specifications (term and default factors, commodity carry and momentum, FX carry and value), which we do not implement.